La Scuola d'Italia / Dr. Huson / IB Math AA SL
13 May 2026

First & Last name:

11.18 Classwork: Series

Problems 1 and 2 are no calculator. Problems 3 and 4 are calculator allowed.

1. [Maximum mark: 5] AA SL 2024 May P1 TZ1 Q6

   Consider a geometric sequence with first term 1 and common ratio 10.

   S_n is the sum of the first n terms of the sequence.

   (a) Find an expression for S_n in the form $\dfrac{a^n - 1}{b}$, where $a, b \in \mathbb{Z}^+$. [1]

   (b) Hence, show that

   $$S_1 + S_2 + S_3 + \cdots + S_n = \frac{10(10^n - 1) - 9n}{81}.$$

   [4]

2. [Maximum mark: 8] AA SL 2025 May P1 TZ1 Q6 (no calculator)

Consider a sequence of ten rectangular picture frames $F_1, F_2, \ldots, F_9, F_{10}$.

Picture frame F_1 has width 4 cm and height 5 cm.

The width and height of picture frame F_n are each increased by 50% to generate the width and height of the next picture frame F_{n+1}, for $n \in \mathbb{Z}^+$, $1 \leq n \leq 9$.

(a) (i) Show that the area of picture frame F_n is $20\left(\frac{9}{4}\right)^{n-1}$ cm^2. [2]

(ii) Hence, find the mean area of the ten picture frames, giving your answer in the form $p\left(\left(\frac{9}{4}\right)^{a}-1\right)$ cm^2, where $p \in \mathbb{Q}^+$ and $a \in \mathbb{Z}^+$. [3]

(b) Find the median area of the ten picture frames, giving your answer in the form $q\left(\frac{9}{4}\right)^{4}$ cm^2, where $q \in \mathbb{Q}^+$. [3]

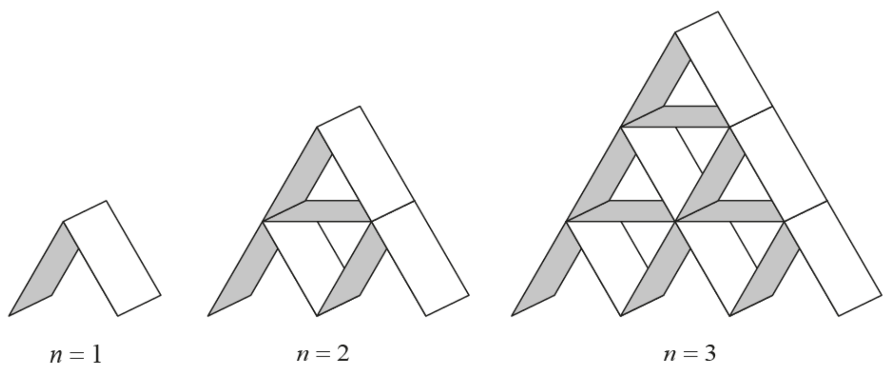
First & Last name:

3. [Maximum mark: 16] AA SL 2025 May P2 TZ1 Q8

Rectangular playing cards are stacked in the shape of a pyramid with n rows, where $n \geq 1$.

Some cards are placed horizontally and some cards are stacked at an angle of 60° to the horizontal.

The following diagrams represent pyramid stacks for $n = 1$, $n = 2$ and $n = 3$.



Let t_n represent the number of cards used to create a pyramid stack with n rows.

(a) Write down t_3. [1]

(b) Find t_4. [2]

(c) Show that [3]

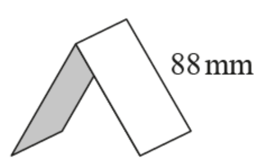
$$t_n = \frac{n(3n+1)}{2}.$$

There are 52 cards in a full pack of playing cards.

(d) A complete pyramid stack is created using playing cards taken from 14 full packs. Find the maximum number of rows in this stack. [3]

(e) A complete pyramid stack is created using playing cards taken from full packs with no cards left over. Find the minimum number of rows in this stack. [2]

The long edge of each playing card measures 88 mm as illustrated in the following diagram.



(f) Find the minimum number of cards needed to create a complete pyramid stack with a vertical height of more than 2 metres. The thickness of the cards may be ignored. [5]

4. [Maximum mark: 12] AA SL 2025 Nov P2 TZ3 Q7

Andy and Jess each have \$5000.

Andy invests the money in a new savings plan that will pay interest at the end of each month.

Andy will receive a fixed amount of interest each month. The amount received is 0.315% of the initial investment.

(a) (i) Determine the amount of interest Andy will receive at the end of each month. Give this answer correct to two decimal places. [1]

(ii) Hence, determine the amount of interest Andy will receive each year. [1]

(iii) Write down an expression in the form $5000 + pn$, where $p \in \mathbb{Z}^+$, for the amount of money Andy will have in his savings plan at the end of n years. [1]

(iv) Hence, show that Andy will have \$5945 in his savings plan at the end of 5 years. [1]

In this part, where appropriate, give all answers to the nearest dollar.

Jess invests her \$5000 in a new account that pays 3% interest compounded annually.

(b) (i) Determine the amount of money that will be in Jess's account at the end of 5 years. [2]

(ii) Hence, find the amount of interest Jess will receive in the 5 years. [1]

(c) (i) Write an expression in the form $5000 \times q^n$, where $q \in \mathbb{R}^+$, for the amount of money that Jess will have in her account at the end of n years. [2]

(ii) Hence, determine the smallest number of complete years that it will take for Jess to have more money in her account than Andy has in his savings plan. [3]